

# Dylan Vidal

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## EDUCATION

### University of Central Florida

Bachelor of Science in Computer Science

Orlando, FL

Aug 2023 – May 2027

## EXPERIENCE

### Software Engineer Intern

May 2026 – Aug 2026

#### NVIDIA

Santa Clara, CA

- Built a distributed pool of reusable rack-scale simulators for NVIDIA systems-software teams, replacing dedicated development hardware with shared infrastructure and reducing reserved inventory costs.
- Designed **4 Kubernetes CRDs** for autoscaling, cutting simulator access time by **99.9%**, from **3.5 hours to 7 seconds**.
- Built VM snapshot restore with secure hashing and dynamic fallbacks, cutting reset time by **98.5%**, from **2h01m to 1m52s**.
- Built browser-based **NVIDIA SSO**, replacing AWS accounts, CLI setup, and kubeconfigs with one login to encourage adoption.
- Provisioned an **Entra OIDC identity provider** and least-privilege RBAC on private **AWS EKS** for reservations, SSH, and SCP.

### President

Feb 2024 – Jan 2026

#### Knight Hacks

Orlando, FL

- Manage a non-profit organization with **650+** members and **\$75,000** dollars in yearly revenue focused on uplifting developers.
- Organized the Knight Hacks VIII Hackathon for over **1024** attendees, with a total of **188** projects submitted.
- Lead a team of **52** students to achieve deliverables such as **60** technical workshops a year and **200+** member mentorship.
- Revamped the mentorship program, expanding the number of participants by **58%** compared to previous school years.
- Personally secured nearly **\$35,000** for the organization by networking with numerous **Fortune 500** companies over 2 years.

### Software Engineer Intern

May 2025 – Aug 2025

#### NVIDIA

Santa Clara, CA

- Contributed to the autonomous hardware recovery team for **NVIDIA Mission Control**, the software layer for AI factories.
- Overhauled **Kubernetes/Helm** for zero-downtime upgrades, eliminating **10 hours/customer** annually and cutting builds **93%**.
- Implemented industry standard horizontal scaling techniques to core services increasing traffic bandwidth.
- Optimized network flow by modifying NGINX ingress patterns and service-to-service communication, reducing network latency.
- Developed a log analysis tool that automatically identifies and filters noisy logs, reducing time-to-debug of core services by **60%**.

### Software Engineer Intern

Dec 2024 – Apr 2025

#### Pheratech Systems

Orlando, FL

- Researched Embodied General Intelligence and Sociobehavioral Robots for crisis and natural disaster search and rescue.
- Developed an internal inventory management system, reducing unnecessary wasteful restock purchases by **40%**.
- Trained custom **YOLOv8** object detection models with a resulting **97%** accuracy in recognizing target objects.
- Aided in securing pre-seed funding by contributing to multiple high-impact technical projects across the company.

## PROJECTS

### ReStory | *Python, PyTorch, Gemini, OpenCV, Raspberry Pi, WebSockets, Docker, RTX 5090*

- Awarded **1st place of 112** projects for **Best Overall** at the SwampHacks XI Hackathon.
- Built an AI wearable for Alzheimer's care using multimodal biometrics to recognize faces and recall conversation history.
- Engineered a streaming pipeline using **ArcFace** and **Gemini 3 Flash** to stream **30FPS** sensor data from a **Raspberry Pi 5**.

### VL-ADK | *Google ADK, Jetson Orin Nano, YoloE, Jetbot, Python, Networking*

- Won **1st of 84** for NVIDIA, **2nd of 142** for Microsoft, and **3rd of 26** for Waymo at ShellHacks 2025.
- Developed a **pseudo-VLA** autonomous system that leveraged an agentic workflow to control a **Jetbot** differential drive robot.
- Implemented an accelerated object detection pipeline using **YoloE** with performance boosted by **CUDA**, decreasing latency **83%**.

### Visuworld AI | *React, Next.js, Tailwind, FastAPI, MongoDB, GLSL, Three.js, Google Gemini*

- Awarded **1st place of 97** projects for the **Best Use of Google Gemini Award** at the Bitcamp 2025 Hackathon.
- Invented a novel pipeline for 3D scene generation by prompting **Google Gemini** to produce **OpenGL GLSL** code.
- Parsed and rendered GLSL shader code in a **React** interface using **Three.js** and **WebGL** to display generated 3D environments.

### Lootcode | *React, Next, Tailwind, SQL, tRPC, TypeScript, Zx, Docker, Linux*

- Awarded **1st place of 14** projects submitted to the Knight Hacks 2024 Spring Semester Project Launch Program.
- Featured on **Linux Magazine for Free Open Source Software** after reaching **50,000** page visits and **500** monthly active users.
- Engineered a secure code-grading server and IDE using isolated **Docker** containers to mitigate security vulnerabilities.

## TECHNICAL SKILLS

**Programming Languages:** Python, C++, C, TypeScript, JavaScript, Java, SQL, C#, HTML, CSS

**Libraries/Frameworks:** React, Next, PyTorch, scikit-learn, pandas, numpy, Tailwind, tRPC, Prisma, Drizzle, NextAuth, Kubernetes, Helm, Qt, LangChain, Gemini, ADK, Pyside6, Selenium, Nextcord, Tkinter, Pygame

**Tools/Platforms:** Linux, Docker, GitHub, Node, Bun, pnpm, Jupyter Notebooks, Raspberry Pi, Windows, MacOS, Coolify, Vercel